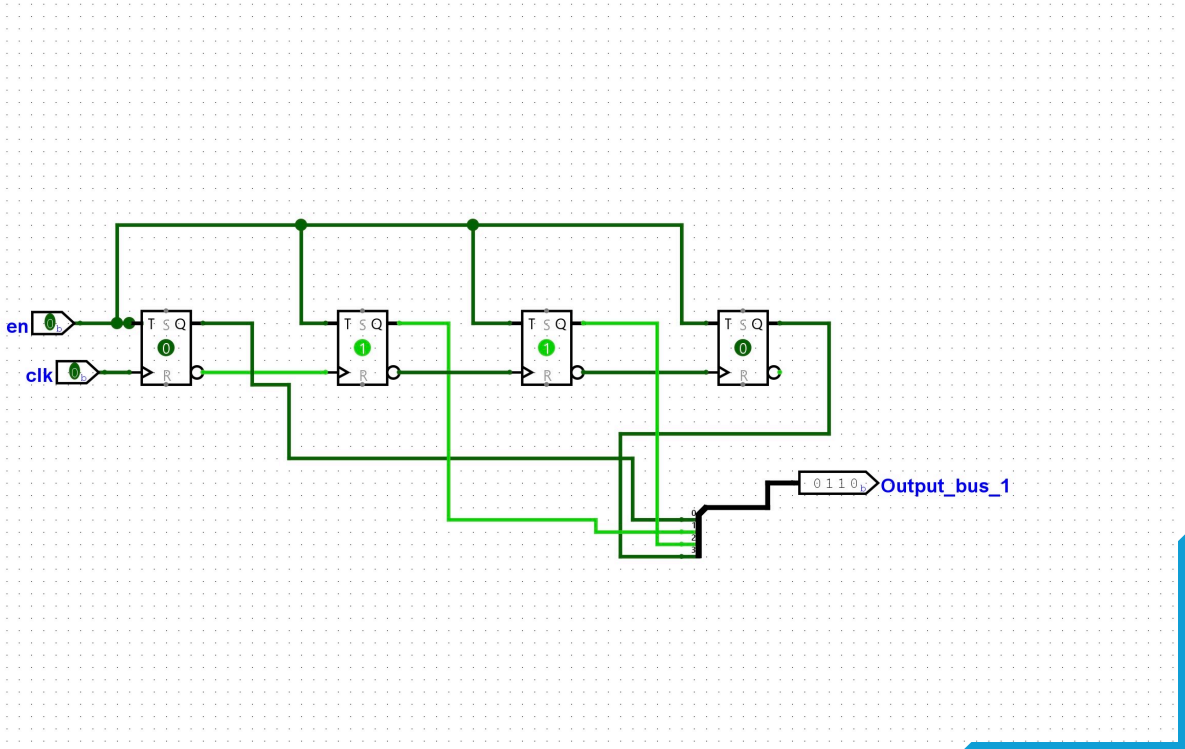
Lab 6

Grace Dreiling

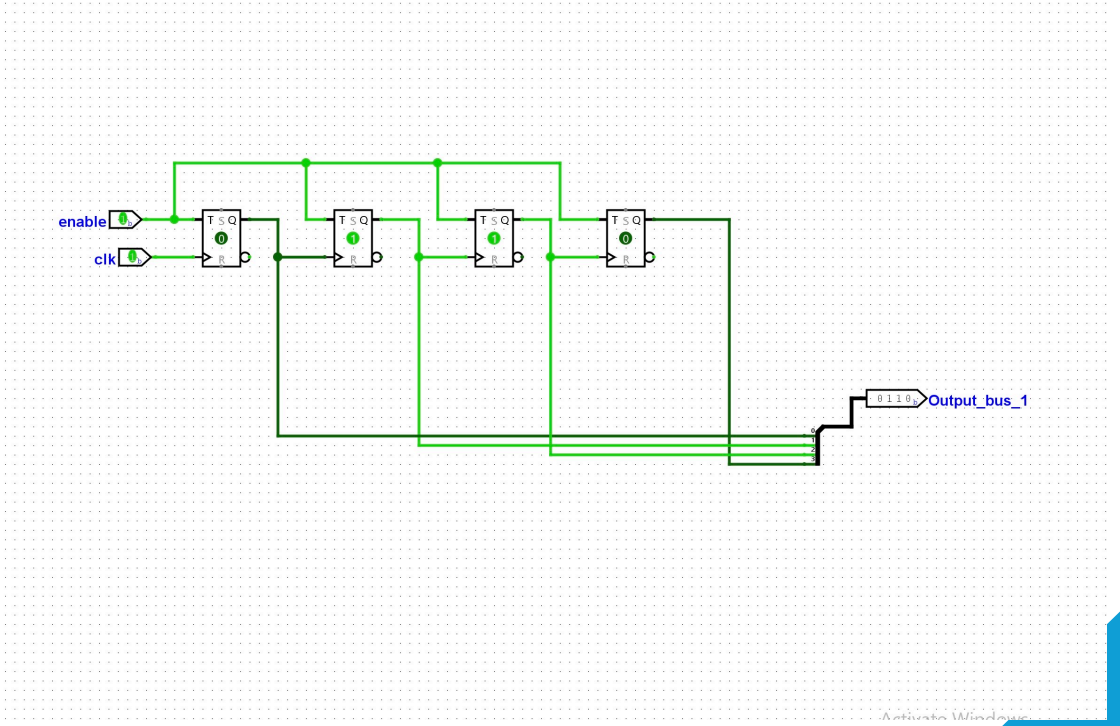
Goal of Lab

Build a cross-walk down counter that goes from 10 to 0

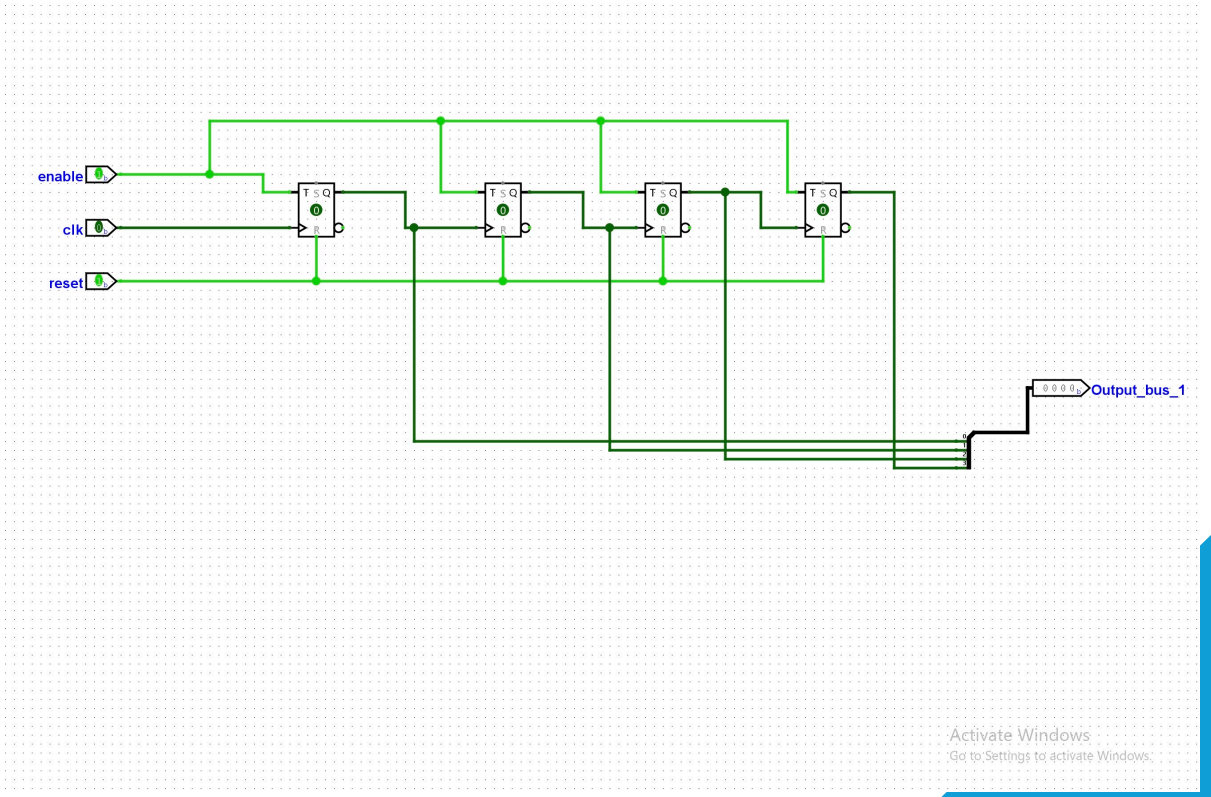
4-Bit Up-Counter



4-Bit Down-Counter



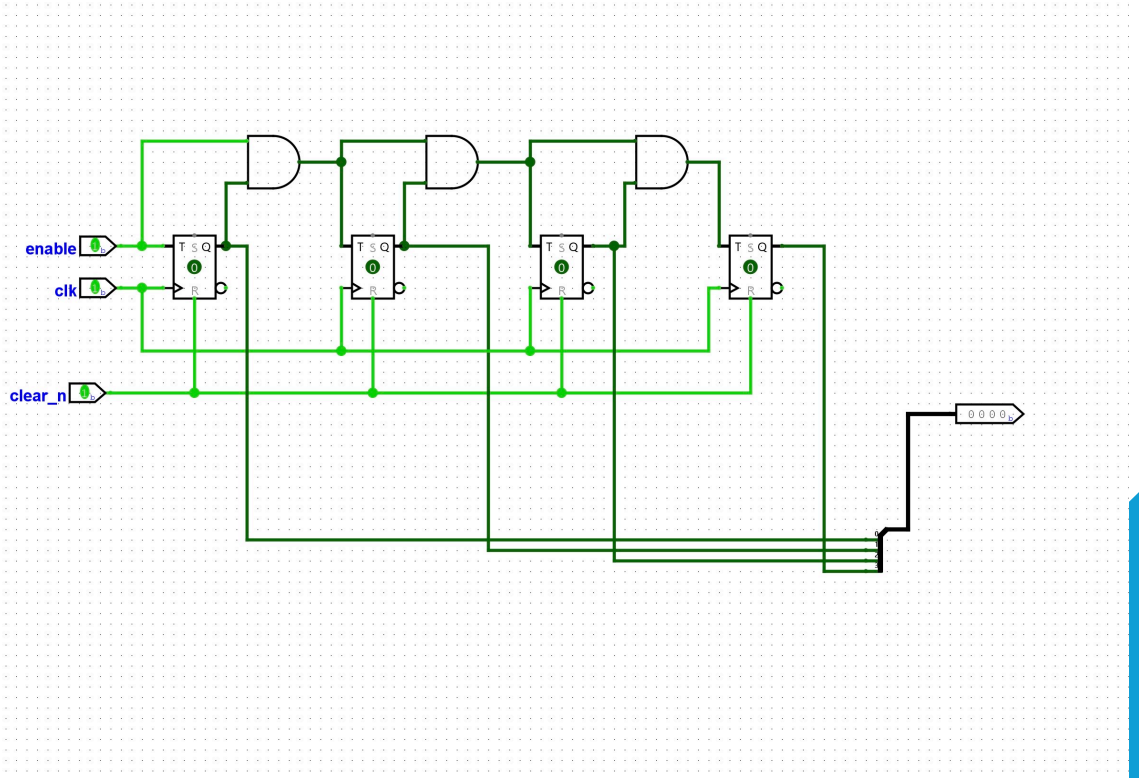
4-Bit Down-Counter with clear



Note: this circuit went from 15-0 and cleared to 0 when button pressed

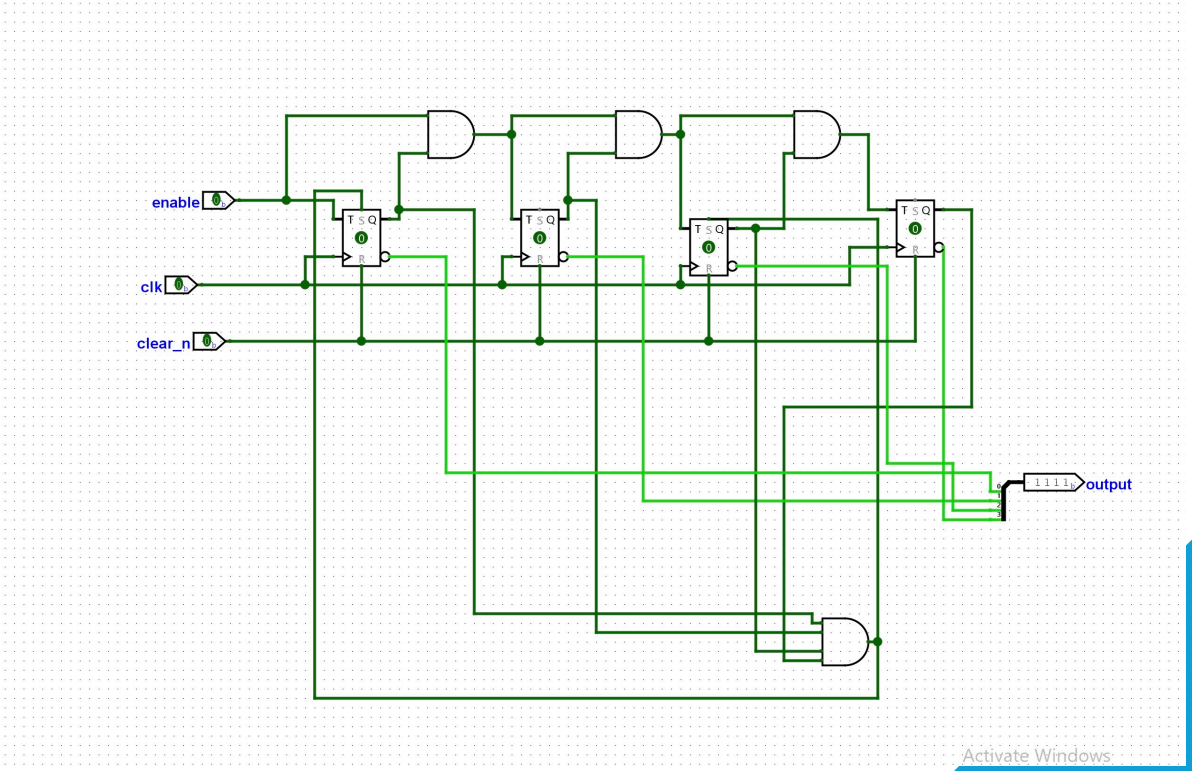
4-Bit Up-Counter with clear

Note: this circuit went from 0-15 and cleared to 0 when button pressed



Synchronous 4-Bit Down-Counter with Reset to 10

Used in the final project



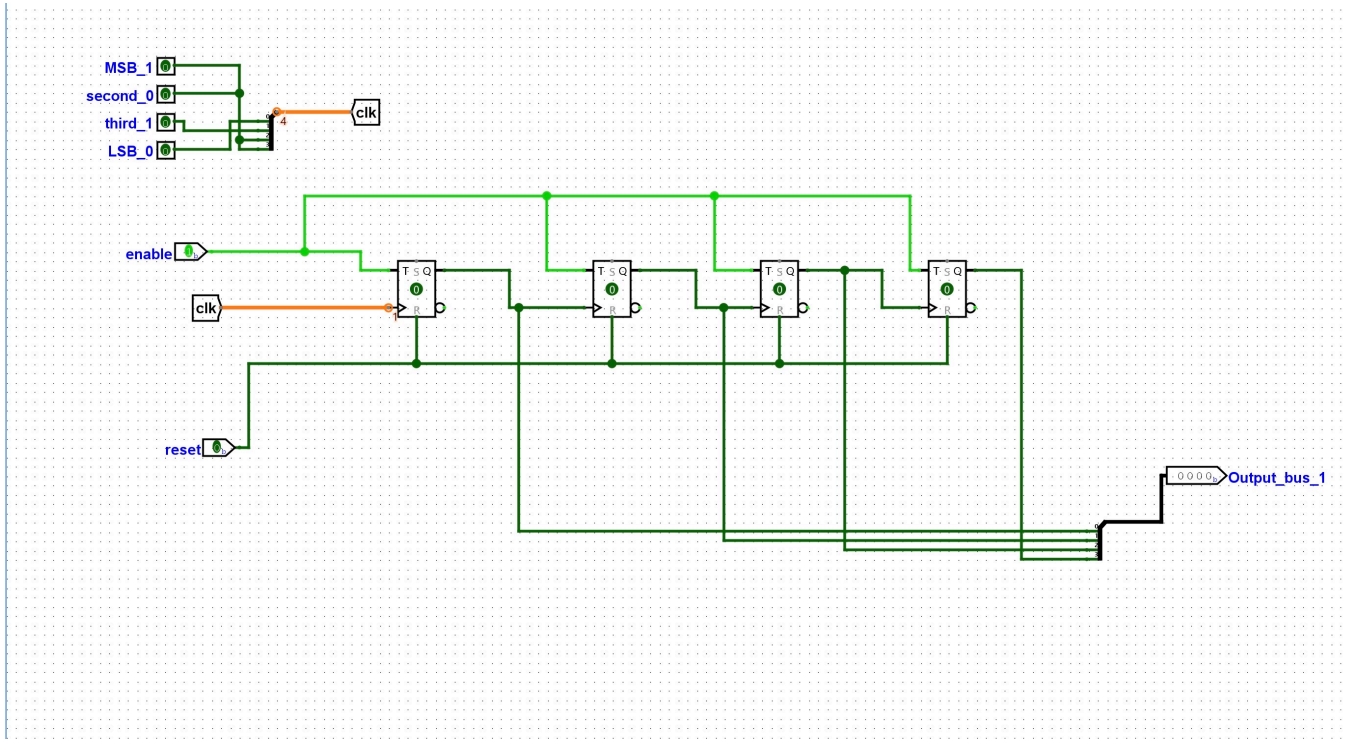
Activate Windows

The Process

Failures: journey to finding the correct circuit

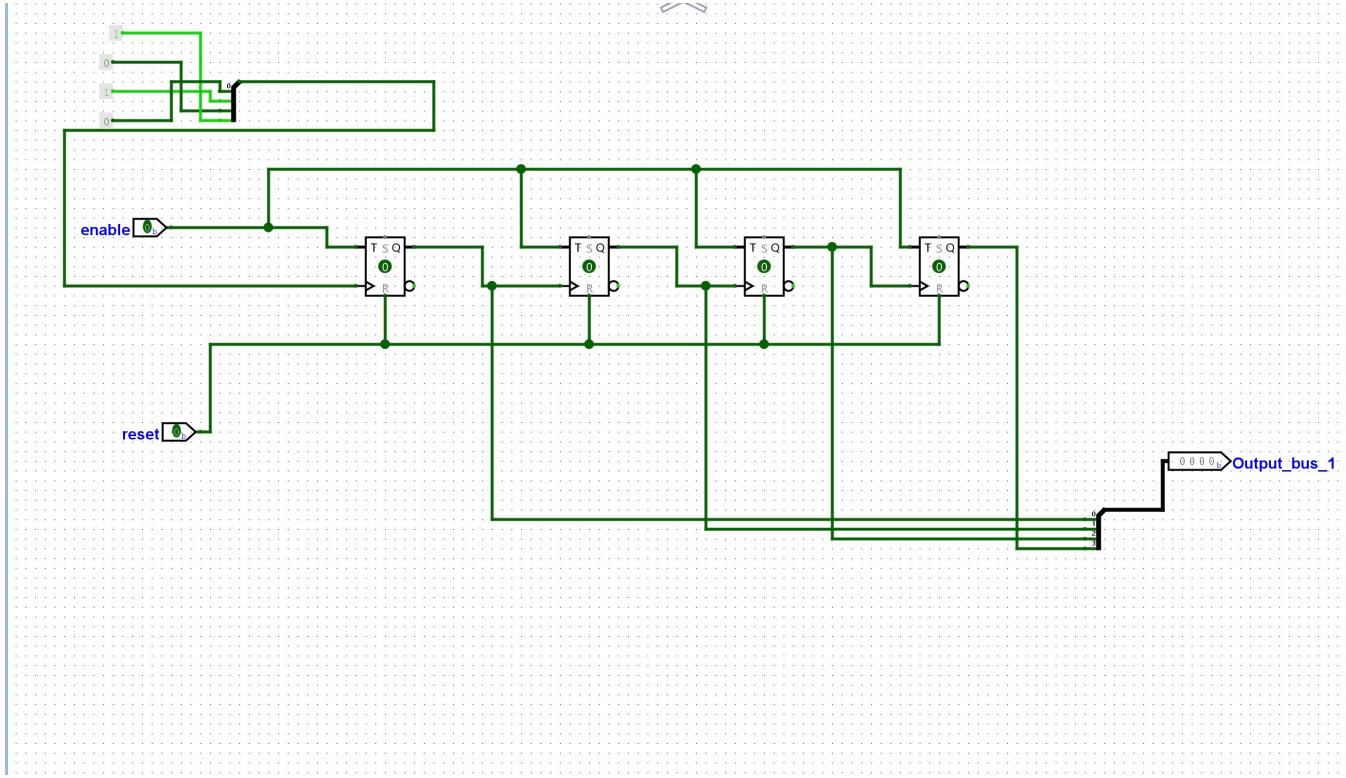
Failed Circuit 1

- Thinking process: how can I have it reset to 10 (1010)? Maybe I can set the clk to that value



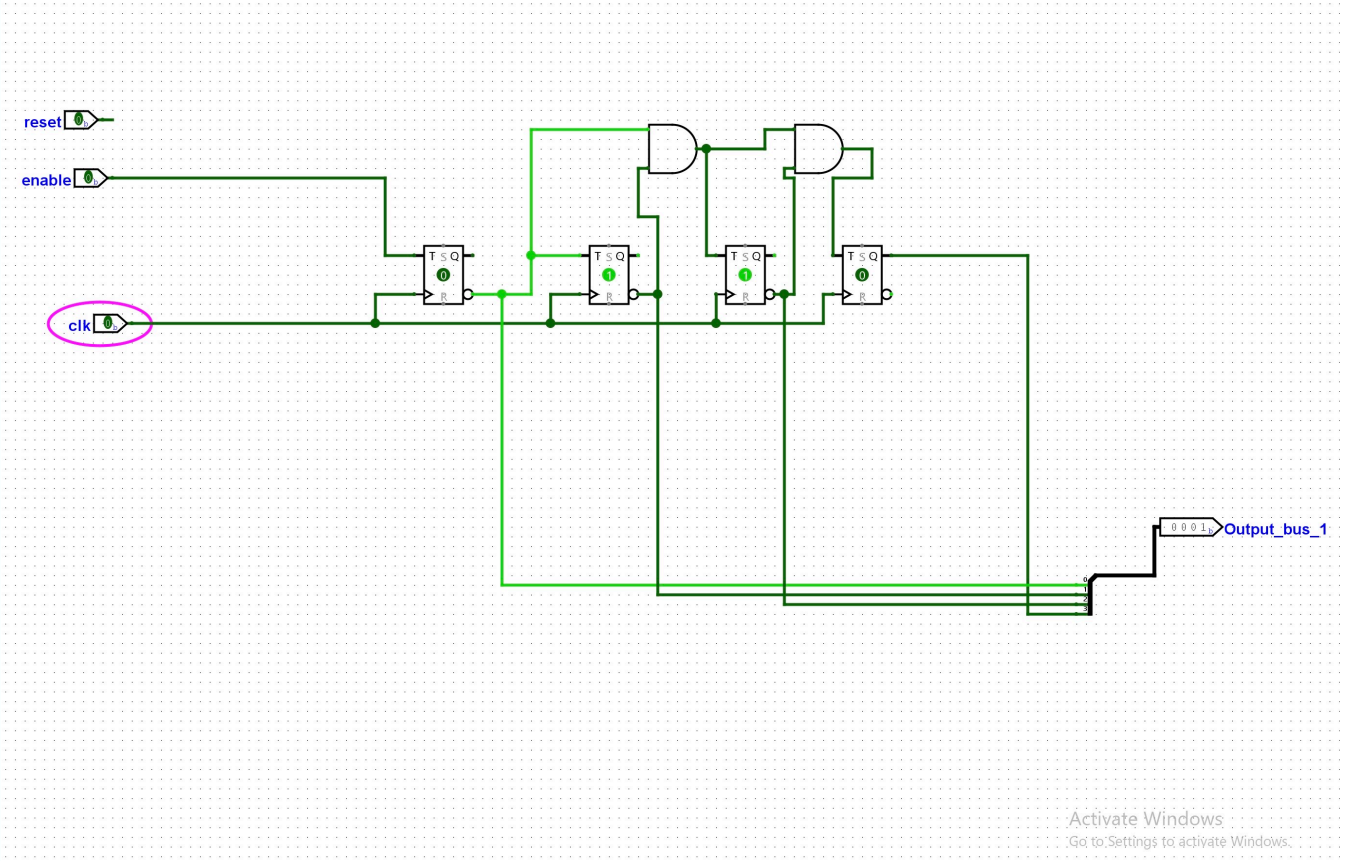
Failed Circuit 2

- Thinking process: how can I have it reset to 10 (1010)? Maybe I can set the clk to that value but try constants since I had a problem with bit width in other attempt



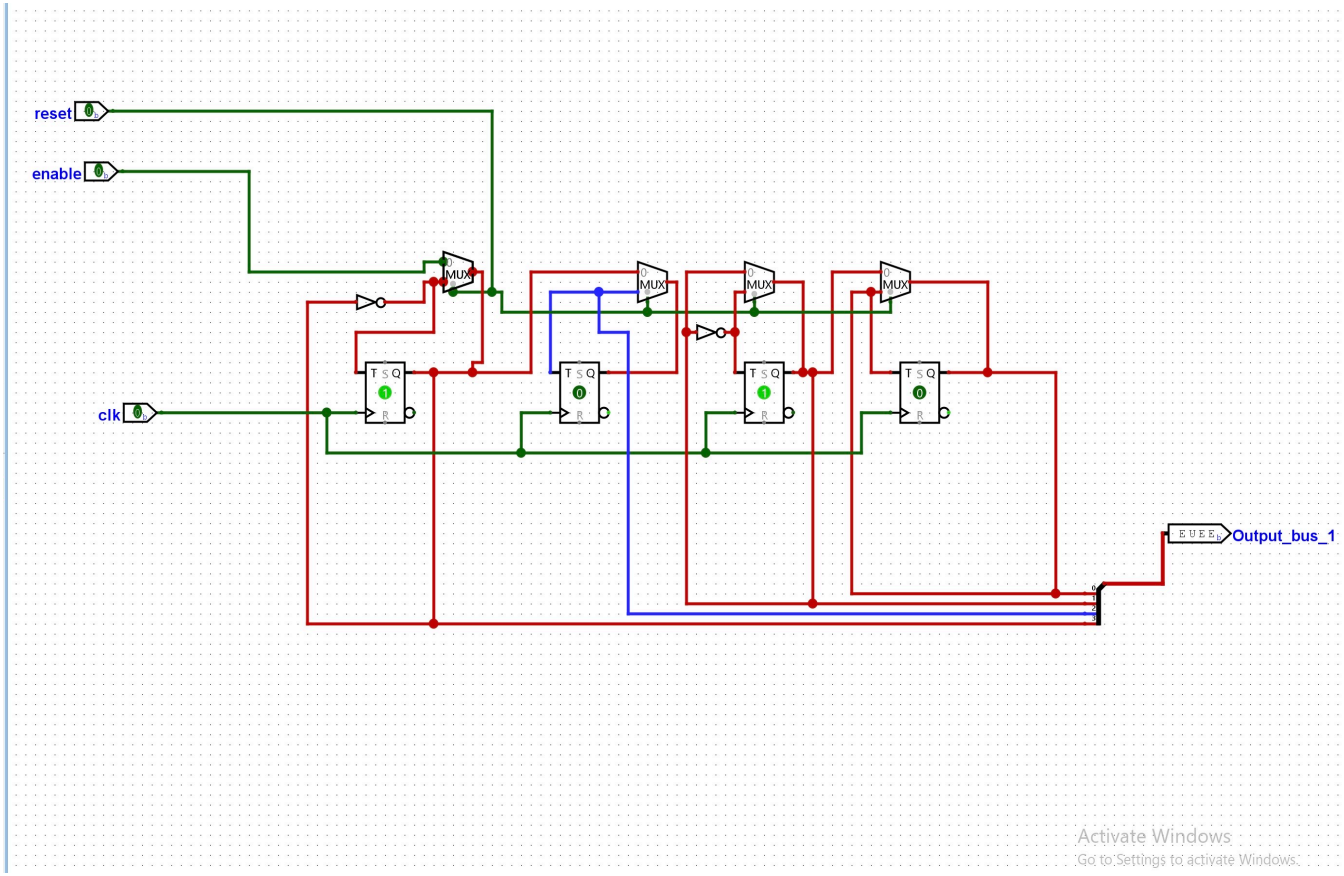
Failed Circuit 3

- Trying to make a synchronous counter (since that was mentioned in class) but getting a little lost in process



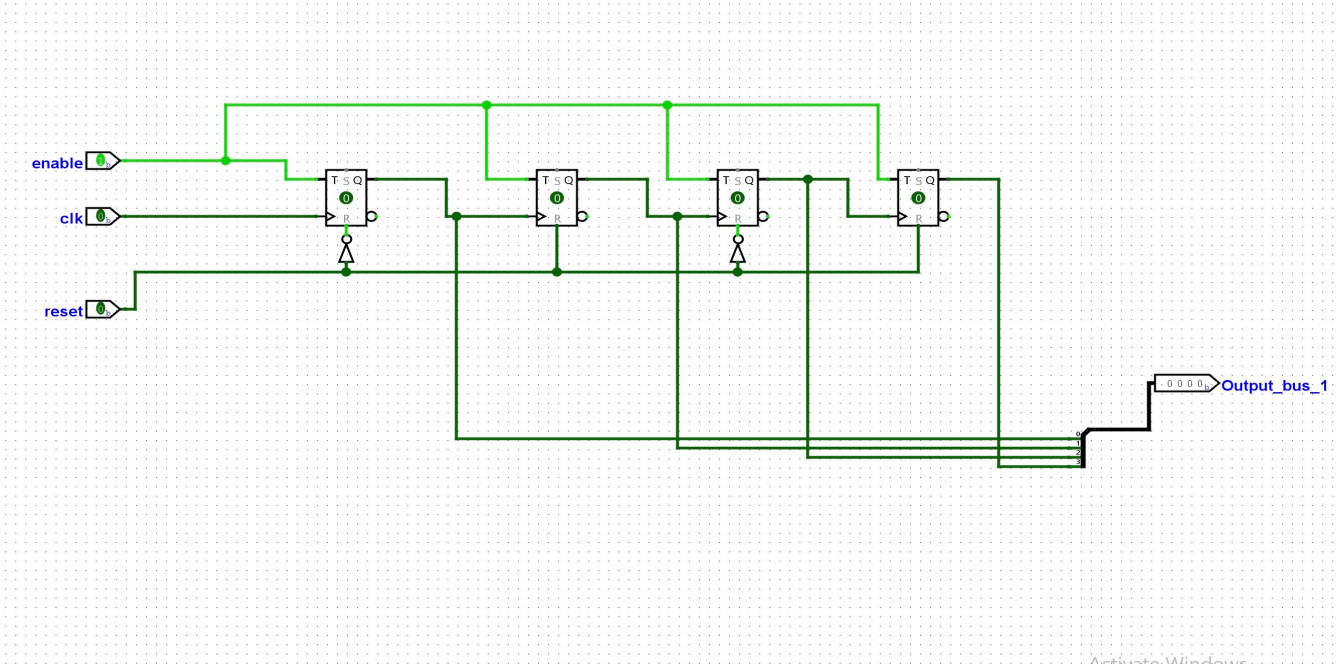
Failed Circuit 4

- Thinking process: I have a down-counter, but how can I get it to set or reset to 10? I'll see what internet has to say. Internet suggested a bunch of stuff that I didn't know how to implement/failed at implementing



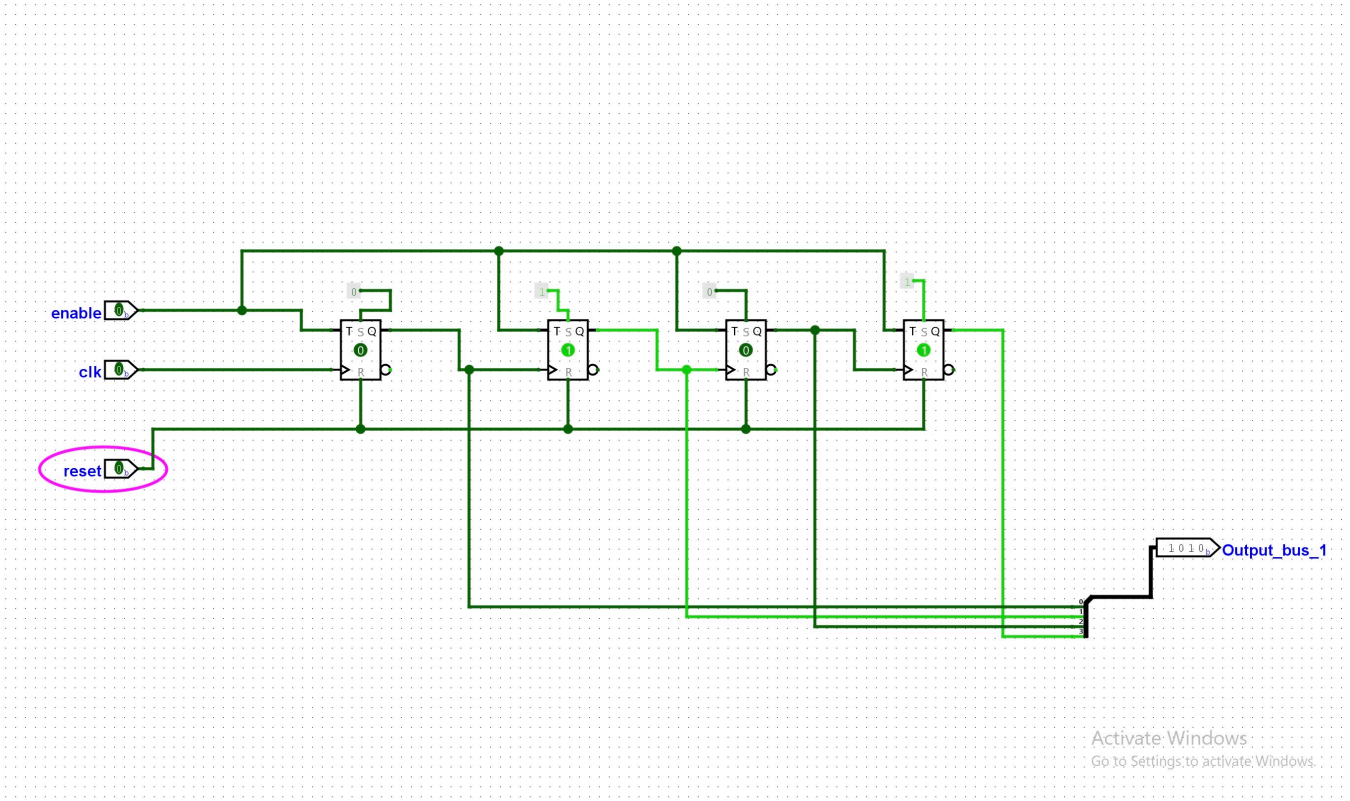
Failed Circuit 5

- Thinking process: What if I connect my reset values with NOT gates. If the reset button puts it all the way to 0, maybe if I use NOT gates, it will put 1s in the places I need to make 10



Failed Circuit 6

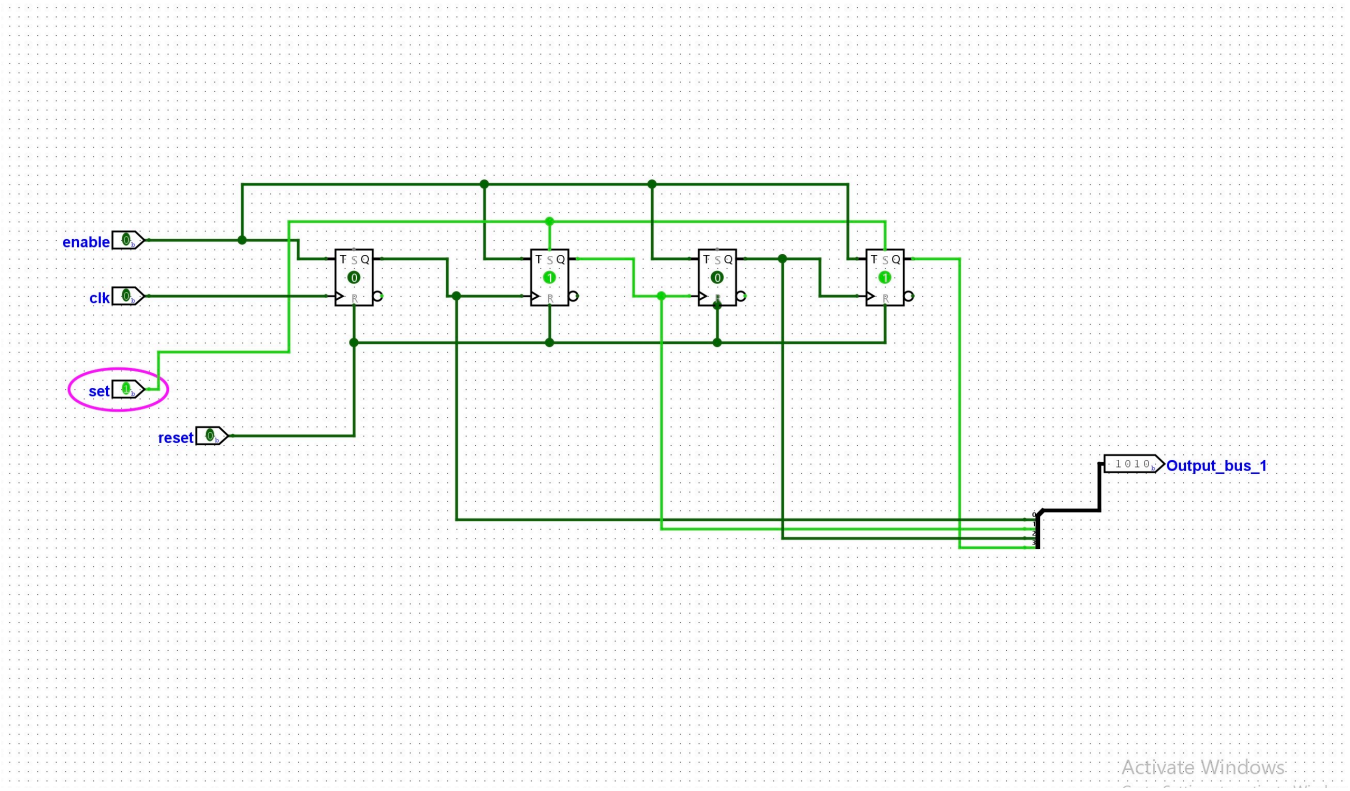
- Thinking process: I realized T-flip flips have set pins, so maybe if I have constant 1s and 0s on right set pins it'll reset/set to 10



Failed Circuit 7

- Thinking Process: What if I just have a button that I call “set” and its only connected to the T-flip flops that I need to set to 1010 to make 10 decimal
- This partially worked
- But it needed me to reset to 0 and set to 10 every time, otherwise it would want to count down from 15 again

Failed Circuit 7

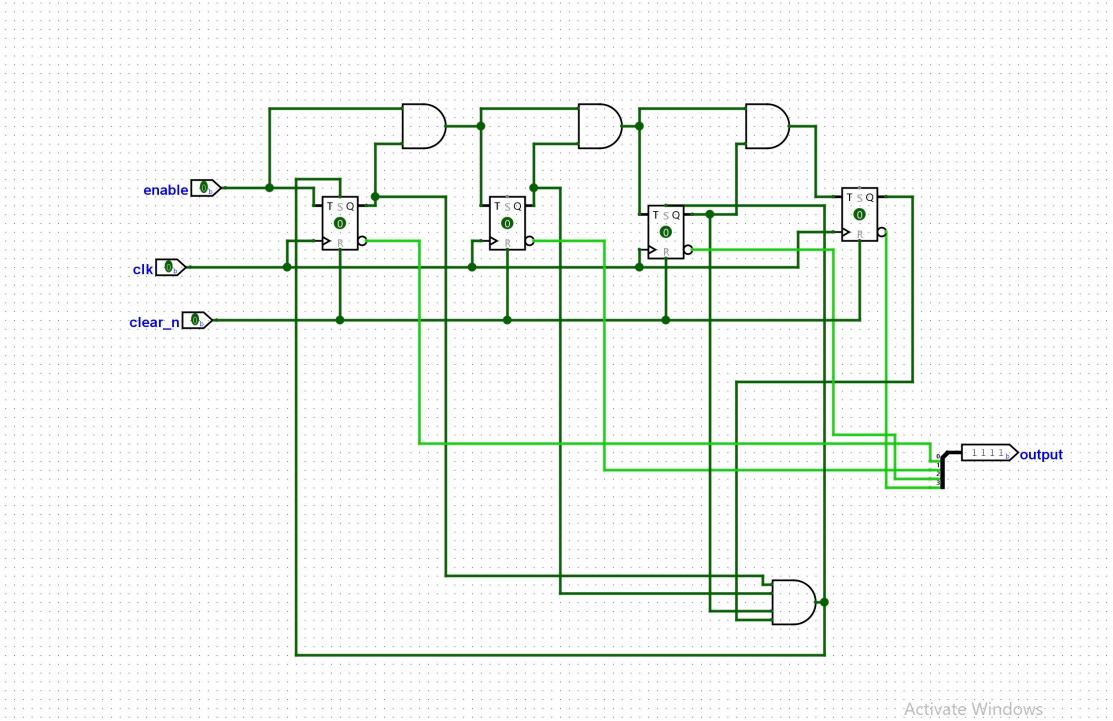


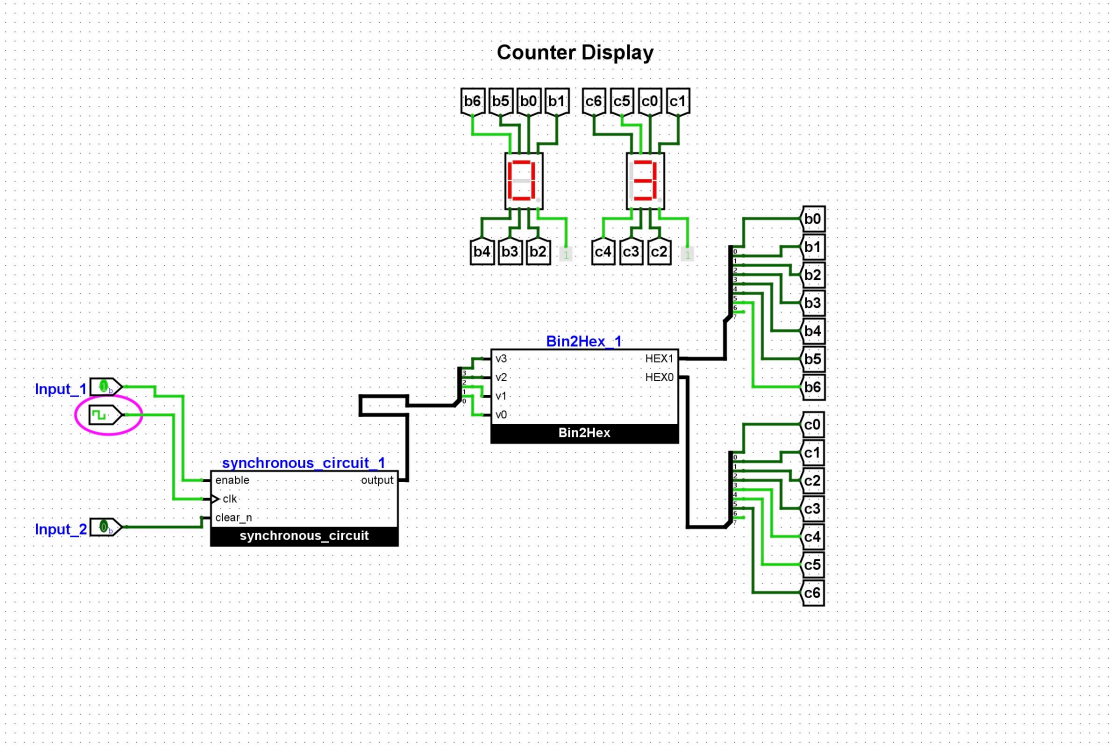
Activate Windows

Final Circuit

The Final Circuit

- Note: the Clear_n button didn’t serve much use, as my synchronous counter was able to naturally reset to 10
- Thinking process: Use the 4-bit synchronous down-counter. Then, how do I get the timer to reset to 10 instead of 15? Need a detection circuit: running the outputs (Q0-Q3) through an AND gate. Then linking the output of the AND gate to the set pins of the T-flip flops that allow for 1010 (10 decimal). When all the outputs are 1 (so 1111 or 15 in decimal), the AND gate produces a 1, which will set the correct pins to 1 to get 10. The circuit detects 1111 (15) and then corrects it to 1010 (10).


enable
clk
clear_n
output
Activate Windows


Counter Display
b6
b5
b0
b1
c6
c5
c0
c1
b4
b3
b2
c4
c3
c2
Bin2Hex_1
v3
v2
v1
v0
HEX1
HEX0
Bin2Hex
Input_1
Input_2
synchronous_circuit_1
enable
clk
clear_n
output
synchronous_circuit
b0
b1
b2
b3
b4
b5
b6
c0
c1
c2
c3
c4
c5
c6

Additional Circuits Used

- 7 segment decoder
- Circuit A and B
- Binary to Hexadecimal (Bin2Hex)
- Comparator